

Name, First name, Student ID _____

Examinant: Prof. Dr.-Ing. Rainer Keller	Number of pages: 10
Courses: Softwaretechnik & Medieninformatik Technische Informatik Ingenieurpädagogik	Semester: SWB4 TIB4 IEP4
Exam: Computerarchitektur	Exam number: IT 105 4003, SWB428
Aids: none, except calculator and 1 DIN A4 paper, both sides written (by yourself)	Duration: 90 minutes

Please read the instructions and questions carefully. Each task has sub-tasks, each of which is assigned a certain number of points. Each point is approximately one minute of time. Do use the available space and time and try to answer as elaborately, yet as thoroughly as possible. You may answer in English or in German.
Please note the **hints and instruction set** on the last pages.

General Questions

(14 Points)

- a) How many Electronic Control units (ECUs, "computers") does a modern car approximately have?

--

1

- b) Name three common characteristics of microcontrollers:

1.	
2.	
3.	

3

- c) Please name 3 upsides and 3 downsides of the C programming language?

Upsides:	
1.	
2.	
3.	
Downsides:	
1.	
2.	
3.	

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- d) In which situations should we use Assembler (ASM), in which rather high-level languages?

In which ASM?	2
In which high-level languages?	

- e) Why is endian-ness relevant when programming in a language like C? Please name two examples of functions:

Relevance endian-ness for high-level language:	3
Examples of functions:	

Architectures

(16 Points)

- a) How is `int x=0xcafe` stored on HCS12, assuming x is located at 0x1022?

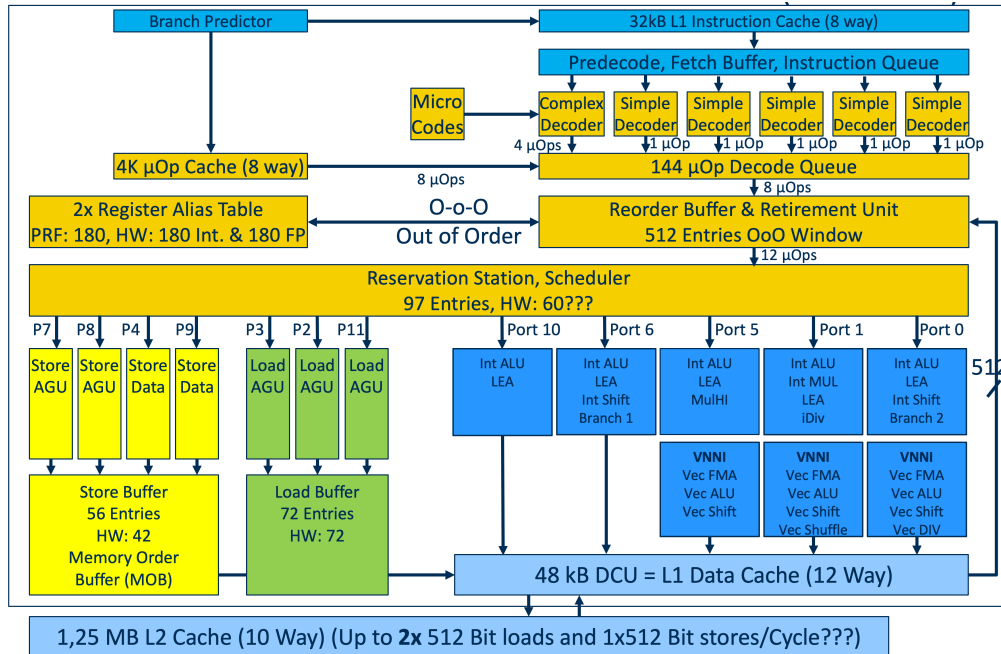
Address	Old Value	Value
0x1020	0x01	
0x1021	0x00	
0x1022	0xff	
0x1023	0x0a	
0x1024	0x00	
0x1025	0x10	

- b) What endianness does Intel / AMD x86-64 provide?

	1
--	---

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- c) Explain in your own words the note-worthy parts of current Intel Alder Lake P-cores – as shown in the architecture diagram below.



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- d) Is the x86-64 architecture **and** specifically the above Intel Alder Lake an Harvard- or a von-Neumann architecture? Please explain your argument?

Harvard- or von-Neumann architecture?

Please explain why:

4

- e) Why is the Apple M2 SoC so performant?

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HCS12 Assembler

(29 Points)

a) How long is shortest & longest HCS12 instruction in binary encoding (in Bytes)?

Shortest:
Longest:

2

b) Please state the error in the following instructions:

Instruction	Error
LDAB #\$cafe	
NEGD	

2

c) Given the following code, state the value of registers D, X and Y after execution:

```
.const: SECTION
        ORG $C000
val1:   DC.B $57, $65, $6C, $6C, $20, $64, $6F, $6E, $65, $00
val2:   DC.W $12, $C00E
val3:   DC.L $89abcdef
.data:  SECTION
        ORG $2000
w:      DS.W 1
```

Instruction	D	X	Y
Initial values:	\$cafe	\$0123	\$abcd
LDAB val1			
LDX val2			
LDY val3			
LDX #val1			
LDD 1, X			
LDY #val2			
LEAY 2, Y			
LDD 0, Y			
LDX -2, Y			
LDX #\$C			
LDD [val1, X]			
LDY 0, Y			
MOVW #AABB, w			
LDX #w			
LDAA +1, X			
LDAB 1, X+			
TFR A, Y			
LDD #\$1122			
LDX #\$3344			
PSHD			
PSHX			
PULB			
PULA			
PULY			

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d) State the value after executing the instructions and the CCR flags?

Instruction	D	Negative (N)	Zero (Z)	Carry (C)
Init: LDD #\$a55a	\$a55a	✓	–	–
ROLA	\$4A5A	0	0	1
ANDB #\$1	\$4A00	0	1	1
ADDB #1	\$4A01	0	0	0
MUL	\$004A	0	0	0
LSRD	\$0025	0	0	0

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The function `f` implements a std. LibC function with 1 parameter.

(Hint: write the results of each line on the right):

`f:`

```

PSHX
PSHY
TFR D, Y    D im Y Register und 0 im X Register
LDX #0

```

L1:

```

LDAB 1, Y+   Wir laden einen 8 bit Wert von der Adresse in Y und Y++
BEQ L2       Wenn der Wert 0 ist.
INX          Wenn Wert in B != 0. X++
BRA L1

```

L2:

```

TFR X, D     Laden X in D
PULY         Laden der Werte aus dem Speicher
PULX
RTS          RETURN

```

e) What is the equivalent std. LibC function; Please state the signature?

stringLength(char * addr)

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f) What's the value in the `B` register when reaching Label L2?

0

2

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g) The function f is inefficient – please optimize by rewriting f:

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h) What does the instruction `BRA *+0` do?

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C Programming

(31 Points)

a) What is the content of variables S1 ... S6 after execution of:

```
char a = 0x1, b = 0xa5, c = 0x10, S1, S2, S3, S4, S5, S6;
S1 = !b;
S2 = b & a;
S3 = b | c;
S4 = b ^ c;
S5 = ~b;
S6 = a >= b ? b : -c;
```

S1=	
S2=	
S3=	
S4=	
S5=	
S6=	

4

b) Consider You have the following function:

```
long f(char * a, int b, long c) {
    int d;
    ...
}
```

Please state, where exactly the data is stored/passed, when using the HCS12 ABI?

	Stored where?	How many bytes?
Parameter a		
Parameter b		
Parameter c		
Variable d		
Return value		

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c) When declaring a C function as Interrupt Service Routine (ISR), how many Bytes have to be saved on the Stack per each invocation of this Interrupt?

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- d) Fill in the following code to use the ADC to measure voltages. The analog input is in the range of 0 ... 5V. First initialize the ADC using the **ATD0** taking **8 samples using 10-bit resolution (right-adjusted)** with the **default 2MHz** sampling clock from **Channel 2**. Convert the value back into a double-digit, fixed-comma value representing the range, i.e. 0 is **0V**, 45 is **4,5V** and 50 represents **5,0V** etc.

```
void initADC(void) {
```

```
    ATD0CTL4 = 0b0000 0101      asm{
                                MOVB #$05,ATD0CTL4
    ATD0CTL3 = 0b0 1000 000
    ATD0CTL5 = 0b100 0 0 010    }
    ATD0CTL2 = 0b1 1 0000 1 1  Required Last due to immediate Interrupt / Start
                                without setting the other registers first
```

4

```
    .....
}
```

```
unsigned int getADC(void) {
```

```
    ATD0CTL5 = 0b100 0 0 010;    1 point
    while((ATD0STAT0 & 0x80) != 0x80); 2 points
    return ATD0DR0                1 point
```

4

```
}
```

```
unsigned int convert(unsigned int measured) {
```

```
    return (measured * 50) / 1023
```

1

```
}
```

```
void main(void) {
    initADC();
    while (1) {
        unsigned int measured = getADC();
        unsigned int converted = convert(measured);
        ...
    }
}
```

- e) What is the highest value `getADC()` may return?

```
1023                [(2^16-1) = max unsigned int value]
```

1

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- f) Measure a signal on Port T.5 using the **Input Capture mode** for Channel 4, triggering on a **both edges**. Set the clock **divider to 0**. Store the current timer value to `lastTime`.

```
void initTimer(void) {
```

<code>TIOS &= 0b 111 0 1111</code>	Sets the Channel into Input Capture Mode
<code>TSCR2 = 0b 0000 0 000</code>	Sets clock divider to 0
<code>TIE = 0b0001 0000</code>	Enables the Interrupts on Channel 4
<code>TCTL3 = 0b 0000 0011</code>	Sets the type of input trigger to dual edge
<code>TSCR1 = 0b 1 000 0000</code>	Starting the Timer

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```
.....  
}  
unsigned int lastTime;
```

```
void timer4Isr(void) {  
    Address of Interrupt : $FFE6, No. 12
```

<code>TFLG1 = 0b 0001 0000</code>	=> Clearing the Interrupt Flag
<code>lastTime = TCNT</code>	
<code>return</code>	

3

```
}  
  
void main(void) {  
    initTimer();  
    while (1) {  
        ... // uses lastTime  
    }  
}
```

Value is `SCI0BD`

- g) Initialize the **Serial Interface 0** to **57600 baud** using **8 data bits, no parity and 1 stop Bit (8N1)**, and **enable receiver and transmitter**, but **without interrupts**.

```
void initSerial(void) {
```

<code>//SCI0CR1 Already correct out of Reset</code>	<code>(SCI0CR1 = 0)</code>
<code>SCI0CR2 = 0b 00 00 11 00</code>	=> Enables Transmitter and Receiver
<code>SCI0BD = 26</code>	<code>BD = 24000000 / (16 x 57600)</code>

3

```
}
```

